AIPI 590 · Responsible AI · Duke University · Fall 2026

MemoryMirror

See, trace, and correct the memories an assistant is using.

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Interactive XAI demo · Project 1



MOTIVATION

Assistants now keep durable "memories" about people. When those memories are wrong, stale, or over-inferred, the person usually cannot see the record, cannot prove where it came from, and cannot test whether fixing it changes the next reply.

RESEARCH QUESTION

When an assistant answers from a stored memory, can a person see that memory, trace it to a source message, correct it, and get a different answer to the same question under the same model and the same agent settings?

Domain: Human–AI interaction · Explainable AI · Memory provenance
Audience: People subject to assistant memory, and builders who need a correction loop they can evaluate

CONTRIBUTION

An interactive memory lab

- Profile of active beliefs
- Lineage graph: message → belief → reply
- Edit / delete with version history kept
- Three-arm counterfactual lab
- 12-case research benchmark + JSON export
- Sensitivity filter; local-first privacy

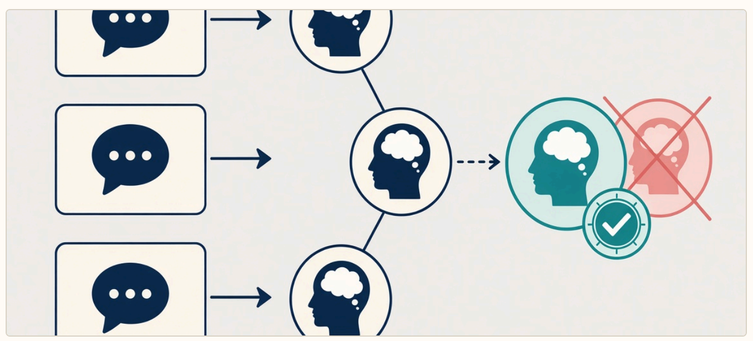


Figure 2 · Messages ground beliefs; corrections supersede without erasing history

METHOD

Hold the model fixed. Change only memory.

Three synthetic people ship with a planted stale or over-inferred belief. The lab freezes model id and agent options, then swaps only the memory snapshot fed to the prompt.

01 See
Read the belief card

02 Trace
Open the source message

03 Correct
New version; old kept

04 Re-ask
Same question, new memory

EVALUATION

Evidence, not vibes

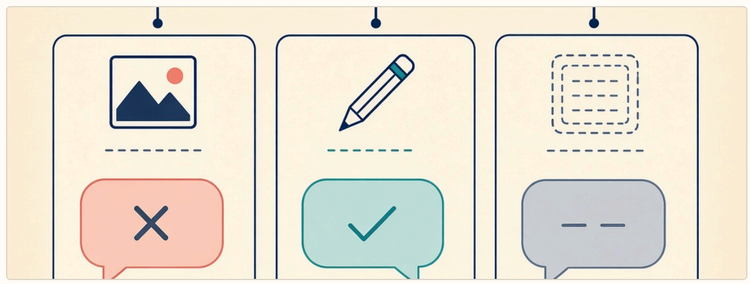


Figure 3 · Same question × original / edited / without-memory arms

Correction adherence
Reply must include corrected facts

Stale-memory leakage
Old facts must not reappear (negation window)

Unsupported inference
Blocks diagnoses the seed never stored

INSIGHTS

- A correction that never reaches the prompt is theater — the reply arm must consume the new snapshot.
- Provenance edges come from recorded ids, not fuzzy wording similarity.
- Fixture mode proves the pipeline; live Cursor runs are the model score.
- Substring rubrics are honest about their limits — and force reproducible claims.

LIMITATIONS & FUTURE WORK

- Synthetic people and planted stale beliefs — not a field study of production chat logs.
- Causal claim is scoped: replies differ because the memory list differed, under fixed agent settings.
- Rubric is lexical; paraphrases can fail adherence or false-flag leakage.
- Next: user study on trust repair; stronger paraphrase scoring; rate limits before wider deploy.

TAKEAWAY

Make the memory loop visible enough to audit — and measurable enough to fail.

Artifact: Next.js demo with fixture + live Cursor agent modes, lineage explorer, counterfactual lab, and a 12-case benchmark you can re-run and export.